



REACTIVE FORM IN ANGULAR



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What Are Reactive Forms?

Reactive Forms are structured, model-based forms in Angular.

- **Built entirely in the TypeScript component**
- **Highly scalable, predictable, and testable**
- **Perfect for complex forms, dynamic validation, and real-time control updates**

✓ Import ReactiveFormsModule

📦 First, enable Reactive Forms:

```
app.component.ts

import { ReactiveFormsModule } from '@angular/forms';

@NgModule({
  imports: [ReactiveFormsModule]
})
export class AppModule {}
```

📦 Required for using FormControl, FormGroup, and FormBuilder.

✓ Create FormGroup in Component

```
app.component.ts

import { FormGroup, FormControl } from '@angular/forms';

form = new FormGroup({
  name: new FormControl(''),
  email: new FormControl(''),
});
```

💡 Each field is explicitly defined in the component, giving you full control over logic, state, and validation.

✓ Bind Form to Template



app.component.html

```
<form [formGroup]="form" (ngSubmit)="submitForm()">
  <input formControlName="name" placeholder="Name" />
  <input formControlName="email" placeholder="Email" />
  <button type="submit">Submit</button>
</form>
```

📌 Angular tracks form state automatically (valid, dirty, touched).

✓ Validation in Reactive Forms

```
app.component.html

import { Validators } from '@angular/forms';

form = new FormGroup({
  name: new FormControl('', Validators.required),
  email: new FormControl('', [Validators.required,
Validators.email]),
});
```

✓ Types of built-in validators:

- required
- minLength, maxLength
- email
- pattern

✓ Custom Validators

You can create reusable logic:

```
app.component.ts

function noSpaceValidator(control: FormControl) {
  return control.value.includes(' ') ? { noSpace: true } : null;
}

new FormControl('', noSpaceValidator);
```

🔥 Custom validators give fine-grained control over business rules.

✓ Dynamic Forms & FormBuilder

Use FormBuilder for cleaner syntax:

```
app.component.ts

import { FormBuilder } from '@angular/forms';

constructor(private fb: FormBuilder) {}

form = this.fb.group({
  username: ['', Validators.required],
  password: ['', [Validators.required, Validators.minLength(6)]],
});
```

FormBuilder makes form creation easier and more readable.

Real-World Features

Reactive Forms support:

- **Dynamic addition/removal of fields**
- **Async validators**
- **Nested FormGroups**
- **Reactive state tracking**
- **Conditional validation**
- **Unit testing**

 **Ideal for complex enterprise forms
(multi-step forms, wizards, etc.)**

Key Takeaways

- **Full control from the component using FormGroup & FormControl**
- **Better suited for dynamic, scalable, and testable forms**
- **Built-in and custom validations supported**
- **Perfect for complex, reactive UI behavior**



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